

Article

Financial Literacy and Impulsivity: Evidence from Japan

Takuya Katauke, Sayaka Fukuda, Mostafa Saidur Rahim Khan * and Yoshihiko Kadoya 

School of Economics, Hiroshima University, 1-2-1 Kagamiyama, Higashihiroshima 7398525, Japan; b195061@hiroshima-u.ac.jp (T.K.); b195789@hiroshima-u.ac.jp (S.F.); ykadoya@hiroshima-u.ac.jp (Y.K.)

* Correspondence: khan@hiroshima-u.ac.jp

Abstract: The existing literature considers financial literacy to be a proxy for rational decision-making instruments. Although there is empirical evidence on the impact of financial literacy on improving rational decision-making ability, it is not yet known whether financial literacy reduces irrational decisions. Impulsive decisions are a form of irrationality where people prefer smaller but earlier rewards over larger but delayed rewards. Thus, impulsive decisions lead to suboptimal decisions in terms of utility gain. This study investigated whether financial literacy reduces impulsivity in financial and economic decisions. We use data from the Preference Parameter Study (PPS) of Osaka University. We measure hyperbolic discounting as a proxy for impulsive decision making. To control for the endogeneity bias between financial literacy and hyperbolic discounting, we use childhood experiences of talking about finances with parents as an instrumental variable. Our probit regression results show that financial literacy is negatively associated with hyperbolic discounting, after controlling for endogeneity bias. Furthermore, we observed that the effect was significant among respondents aged over 40 and among female respondents. Our results suggest that authorities should consider using financial literacy as an alternative policy intervention to change impulsivity preferences.

Keywords: financial literacy; hyperbolic discounting; impulsivity; Japanese survey



Citation: Katauke, T.; Fukuda, S.; Khan, M.S.R.; Kadoya, Y. Financial Literacy and Impulsivity: Evidence from Japan. *Sustainability* **2023**, *15*, 7267. <https://doi.org/10.3390/su15097267>

Academic Editor: Nuno Crespo

Received: 22 February 2023

Revised: 22 April 2023

Accepted: 24 April 2023

Published: 27 April 2023



Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

This study investigates the influence of financial literacy on impulsive decisions by using hyperbolic discounting as a proxy. The literature considers financial literacy as a proxy for rational decision-making instruments and substantiates its influence on financial and positive health behavior [1–6]. In contrast, the concept of time discounting has been established as a measure of irrational decisions, such as impulsiveness [7,8]. Previous studies also empirically showed the impact of time discounting or impulsiveness on risky health behavior [8,9]. Although financial literacy as a proxy for rational decision-making instruments is theoretically justified [10,11], its association with impulsive decisions has still not been authenticated. Thus, it is still unknown whether rational decision-making instruments reduce people's irrational behavior. Studies by Ottaviani and Vandone [12] and Hastings and Mitchell [13] provide evidence that financial literacy and impulsiveness could coexist significantly in the financial decision-making process and that impulsiveness could reduce the impact of financial literacy in attaining long-term financial benefits. However, a direct measure of the association between financial literacy and impulsiveness is lacking. We hypothesize that financial literacy is negatively associated with hyperbolic discounting, meaning that financially literate individuals tend to be less exposed to impulsive decisions. We argue that because rational investors correctly assess the value of long-term investment [1–3,14,15], they are less likely to suffer from inconsistencies in valuing costs and benefits over shorter and longer horizons.

As a proxy for rational decision-making instruments, financial literacy enables people to understand the value of money and maximize the benefits of money used over a longer period of time [16]. Financial literacy is considered a rational decision-making instrument

because people benefit from financial literacy in optimizing the life-cycle benefits from consumption and savings. For example, Delavande et al. [11] developed a two-period model of savings and portfolio allocation, allowing the acquisition of human capital in the form of financial knowledge. The study of Delavande et al. [11] implies that more financial knowledge is associated with people's access to assets with higher returns. The theoretical models of Hsu [17] and Jappelli and Padula [18] also confirm that financial literacy and wealth are correlated over the life cycle. Lusardi and Mitchell [10] further improved the model by incorporating borrowing restrictions and showed that people would continue to acquire financial knowledge until their marginal benefits equated with money and time costs. All these models also suggest that the need for financial literacy depends on people's life-cycle income and cost profiles. However, despite people's own incentive to invest in financial literacy, there is a socially optimal level of financial literacy because of its benefits [10]. The importance of financial literacy, as laid down in theoretical models, has been substantiated globally. Research findings show that the impact of financial literacy has not only been limited to financial and economic behavior [3,19–22] but has also been extended to risky health behaviors [1,2,5,6]. Some studies claim that financial literacy is associated with improved cognitive performance and helps individuals make time-consistent decisions [1,2,5,6].

The concept of time discounting is frequently used in behavioral economics as a measure of irrationality and limited rationality, which are also known as impulsive decisions [7,8]. Previous studies have shown that hyperbolic discounting, which focuses on the difference in the preference for time over shorter and longer horizons, can capture impulsivity in economic and health-related behavior [8,9,23,24]. However, recent literature indicates that the impatience or impulsivity of financially literate people could have affected the traditional influence of financial literacy on financial behavior [12]. Although financially literate people are assumed to be rational, whether they suffer from impatience or impulsivity is a matter of great importance because the influence of financial literacy on economic outcomes could be reduced by the influence of impulsivity or impatience. Ottaviani and Vandone [12] argue that although financial literacy and impulsivity both affect financial decisions, impulsivity fully mediates the influence of financial literacy on financial decisions. Hastings and Mitchell [13] also found similar evidence, showing that impatience or present bias could produce suboptimal financial decisions even in the presence of financial literacy. Thus, it seems that financial literacy is important but not sufficient to capture a complete perspective of individual decision-making behavior. The role of financial literacy must be studied along with behavioral traits such as impatience and impulsivity.

In the context of a lack of empirical evidence on how financial literacy is associated with impulsivity, we investigated the association between hyperbolic discounting as a proxy of impulsivity and financial literacy after controlling for demographic, socioeconomic, and psychological factors. Our study is related to research on financial literacy, risk preference, and time preference. Although there are no studies directly comparable to ours, Mudzingiri et al. [25] and Mudzingiri [26] examined the influence of financial literacy on the risk and time preferences of students, and they are of some relevance. However, the studies of Mudzingiri et al. [25] and Mudzingiri [26] suffered from a sampling bias, as they were conducted on 192 university students with a Bachelor of Commerce background. In contrast, our study sample is more inclusive and comprehensive and applies a more holistic methodological approach to find the association between financial literacy and impulsivity.

Overall, our study finds a negative association between financial literacy and impulsivity, meaning that financially literate people are less likely to take impulsive decisions. The findings were more pronounced among respondents aged over 40 and among female respondents. Our findings provide empirical support for reducing irrational decisions in the financial marketplace by improving financial literacy. Our study contributes significantly to the existing literature in at least two ways. First, to the best of our knowledge, this is the first study in Japan to examine the effect of financial literacy on hyperbolic discounting

as a proxy for impulsive decisions by using nationwide survey data. Second, our study provides a new perspective on the role of financial literacy in people's economic, social, and health-related behaviors. This article will help achieve economic and financial sustainability in a broader sense. Financial literacy and impulsivity both have associations with the quality of financial and economic decisions. Higher financial literacy and lower impulsivity lead people to make optimal decisions that maximize utility at the personal level and ensure the sustainable allocation of resources at the macro level. Impulsivity or present bias is a tendency to prefer a lower short-term gain over a higher long-term gain. This tendency often causes suboptimal savings and investment decisions and over-indebtedness. The findings of our study will provide insights helpful for achieving financial and economic sustainability in a country.

The rest of the article is structured as follows. Section 2 shows the data and methodology of the study, Section 3 presents empirical results, Section 4 discusses the results, and Section 5 concludes.

2. Data and Methods

The data used in this study were obtained from the 2010 and 2013 waves of the PPS, which has been conducted annually since 2003 by Osaka University's Institute of Social and Economic Research. In the current study, we mainly utilized the sample from the 2013 wave to extract data for the discount rate and other control variables. In addition, we merged some variables related to financial literacy, financial education, and years of education from the 2010 wave, which were not available in the 2013 wave. After merging two datasets and excluding missing data for demographic and socioeconomic factors, our final sample comprised 1850 individuals, accounting for 42.62% of valid respondents in 2013 (the effective collection number was 4341 individuals).

Our dependent variable, hyperbolic discounting, was measured by responses to the intertemporal questions (Appendix A), following previous literature [24]. We asked the respondents to choose between two options "A" and "B" with alternative interest rates from low to high. To elicit hyperbolic discounting, we used questions Q3 and Q4 in Appendix A, controlling for the length of the delay of seven days at present and after three months. After calculating the two-time discount rates of DR1 and DR2 for Q3 and Q4, we coded 1 for hyperbolic discounters if $DR1 > DR2$ and 0 otherwise.

Our main explanatory variable, financial literacy, was measured using three questions in (Appendix B) following some existing literature [10,19,27–32]. We assigned one point for each correct answer and zero points for each incorrect answer. We then calculated the average score to create a financial literacy variable.

Furthermore, we added variables related to financial education as independent variables. For financial education, we created a binary variable from the question, 'Did you receive any compulsory financial education when you were in elementary school?' with three possible responses: yes, no, or do not know. We coded 1 for respondents who responded "yes" and for respondents who answered "no" and "do not know" as zero. In addition, demographic and socioeconomic variables were included as control variables. These include gender, age, marital status, university degree, household number, employment status, household income, and household assets. Moreover, we added the respondents' risk aversion measured by the statement "How high does the chance of rain have to be before you will bring an umbrella with you when you go out?"

The variable definitions are shown in Table 1.

2.1. Descriptive Statistics

The descriptive statistics of the sample are presented in Table 2. Approximately 39.1% of respondents indicated a tendency for hyperbolic discounting. For financial literacy, on average, respondents' financial literacy scores were 0.63, while only 14.4% of the sample received financial education when they were in elementary school. Regarding demographic characteristics, 47.4% were male, and the average age was 50.0 years old. A total of 30.4% of

the sample had graduated from university. Approximately 83.4% were currently married and had at least 1 child, while only 2.5% had divorced. A total of 72.8% had a current job. Regarding socioeconomic factors, respondents had around JPY 6.5 million in annual household income and JPY 14.7 million in household financial assets in the 2013 wave. Finally, on average, 46.6 % of respondents showed a tendency toward risk aversion.

Table 1. The definition of variables.

Variables	Definition
Hyperbolic discounting	A binary indicator for hyperbolic discounting which equals one if $DR1 > DR2$, and zero otherwise.
Financial Literacy	Continuous variable: average score for the number of current answers from three financial literacy questions
Financial Education	Binary variable: 1 = received compulsory financial education at school 0 = otherwise
Male	Binary variable: 1 = male, 0 = female
Age	Age of participants
University degree	Binary variable: 1 = obtained a university degree or higher, 0 = otherwise
Marriage	Binary variable: 1 = married, 0 = otherwise
Divorce	Binary variable: 1 = divorced, 0 = otherwise
Unemployed	Binary variable: 1 = unemployed, 0 = otherwise
Household income	Annual earned income before taxes and with bonuses of the entire household in 2009 (unit: JPY million)
Household assets	Balanced amount of financial assets (savings, stocks, insurance, etc.) of entire household (unit: JPY million)
Household Number	The number of people living in the household
Children	Binary variable: 1 = have at least one child, 0 = otherwise
Risk Aversion	A variable that measures the degree of risk aversion from the question: ‘Usually when you go out, how high does the probability of rainfall have to be before you take an umbrella?’
Instrumental variables	Definition
Talk About Finance	Instrumental variables of financial literacy: 5-scale measure from the following question “When I was a child, my parents often talked to me about finance.”, with responses from 1 as “it is particularly true” to 5 as “it doesn’t hold true at all”.

The distributions of hyperbolic discounting, classified by age group, are described in Table 3. There was no statistically significant difference in hyperbolic discounting between generations. This finding implies that age does not have a strong influence on impulsivity. Even so, it is combined with the effects of other demographic characteristics.

The distribution of hyperbolic discounting, grouped by demographic characteristics, is presented in Table 4. We can see significant differences in hyperbolic discounting related to certain characteristics, such as gender, employment status, and marital status. Approximately 41.5% of male respondents were hyperbolic discounters, while for females, the ratio was 36.9%. The proportion of employed respondents among hyperbolic discounters (40.09%) was higher than that of unemployed respondents (36.38%).

Table 2. Descriptive statistics.

Variable	Mean	Std. Dev.	Min	Max
hyperbolic discounting	0.3908108	0.488064	0	1
financial_literacy	0.6261261	0.3264694	0	1
financial_education	0.1443243	0.3515133	0	1
Male	0.4740541	0.4994614	0	1
Age	49.9973	12.65537	21	77
age_squared	2659.802	1265.698	441	5929
Married	0.8356757	0.3706698	0	1
Divorce	0.0254054	0.1573956	0	1
University degree	0.3043243	0.4602451	0	1
household_number	3.514595	1.415763	1	9
Children	0.8335135	0.3726175	0	1
Unemployed	0.2718919	0.4450548	0	1
log_of_household income	15.51089	0.601635	13.81551	16.81124
log_of_household assets	15.93853	1.017686	14.7318	18.42068
risk_aversion	0.4658703	0.1960089	0	1
Observation	1850			

Table 3. Distribution of hyperbolic discounting by age group.

Hyperbolic Discounting	Age			Total
	≤40	41–60	≥60	
0	261	554	312	1127
	63.97%	58.69%	62.65%	60.92%
1	147	190	186	723
	36.03%	20.13%	37.35%	39.08%
Total	408	944	498	1850
	100%	100%	100%	100%
F-statistics	F = 2.10			

Table 4. Distribution of hyperbolic discounting by demographic and socioeconomic variables.

Hyperbolic Discounting	Gender		Employed		Married	
	Female	Male	Yes	No	Yes	No
0	614	513	807	320	213	914
	63.10%	58.49%	59.91%	63.62%	70.07%	59.12%
1	359	364	540	183	91	632
	36.90%	41.51%	40.09%	36.38%	29.93%	40.88%
Total	973	877	1347	503	304	1546
	100%	100%	100%	100%	100%	100%
Mean Difference	$t = -2.0298^{***}$		$t = 1.4541^{***}$		$t = -3.5859^{***}$	

Note: *** $p < 0.01$.

Regarding marital status, 40.9% of married individuals were hyperbolic discounters, compared to 29.9% of unmarried individuals.

2.2. Methodology

Using the concept of hyperbolic discounting, we investigate whether financial literacy can be a proxy for rationality in terms of impulsivity. We hypothesized that financially literate people are not impulsive and exhibit time-consistent behavior. Specifically, they choose a larger-later reward rather than an immediate but smaller reward, following their rationality.

As Samuelson [33] shows, economic theories assume that people's discount rates are constant. Technically, an outcome that has utility A if received immediately ($t = 0$) is valued at $A \cdot \delta^t$ if t periods are delayed in the future. Thus, the present time value (V) of receiving (A) at time (t) is given by:

$$V(A, t) = A \cdot \delta^t$$

The discount rate, δ , represents the constant proportional decrease in value with each added delay period [33].

However, in the field of psychology, it has been proposed that humans have a propensity for hyperbolic discounting [34]. This means that humans tend to violate the exponential assumption of a constant proportional discount factor, whereby they tend to discount rewards in the immediate future more sharply than in the distant future [33]. If the respondents are hyperbolic discounters, the shape of the discount function is described as:

$$V(A, t) = A \cdot \frac{1}{1 + k \cdot t}$$

(k is a parameter indicating the rate at which the value is discounted).

In this study, we created a binary variable indicating whether respondents had a tendency for hyperbolic discounting. Because the dependent variable was a binary variable, we employed probit regression in this study. The following equation was estimated:

$$Y_i = f(FL_i, X_i, \varepsilon_i)$$

where Y indicates whether i respondents are hyperbolic discounters, FL is the average financial literacy score, X is a vector of individual characteristics, and ε is the error term.

However, both hyperbolic discounting and financial literacy affect rational and irrational behavior [7,8,34,35]. Therefore, an endogeneity bias can be caused by including financial literacy as an explanatory variable.

To avoid this issue, we used the instrumental variable (IV) method in our analyses. Some previous studies have found that financial knowledge and the ability to deal with financial issues are influenced by parental attitudes toward finances [29,36–38]. Thus, we used the variable indicating how much respondents talked with parents about finances when they were children as the IV in this study.

Our full estimated equation is as follows:

$$\begin{aligned} \text{Hyperbolic discounting}_i (1 = \text{hyperbolic discounter and } 0 = \text{otherwise}) \\ = \beta_0 + \beta_1 \text{financial literacy}_i + \beta_2 \text{financial education}_i + \beta_3 \text{male}_i + \beta_4 \text{age}_i \\ + \beta_5 \text{married}_i + \beta_6 \text{divorce}_i + \beta_7 \text{uni_degree}_i + \beta_8 \text{household_num}_i + \beta_9 \text{children}_i \\ + \beta_{10} \text{unemployed}_i + \beta_{11} \log_of_hincome_i + \beta_{12} \log_of_hasset_i + \beta_{13} \text{risk_aversion}_i + \varepsilon_i \end{aligned}$$

$$\begin{aligned} \text{Hyperbolic discounting}_i (1 = \text{hyperbolic discounter and } 0 = \text{otherwise}) \\ = \beta_0 + \beta_1 \text{financial literacy}_i (\text{IV} \\ = \text{talk about finance}) + \beta_2 \text{financial education}_i + \beta_3 \text{male}_i + \beta_4 \text{age}_i + \beta_5 \text{married}_i \\ + \beta_6 \text{divorce}_i + \beta_7 \text{uni_degree}_i + \beta_8 \text{household_num}_i + \beta_9 \text{children}_i + \beta_{10} \text{unemployed}_i \\ + \beta_{11} \log_of_hincome_i + \beta_{12} \log_of_hasset_i + \beta_{13} \text{risk_aversion}_i + \varepsilon_i \end{aligned}$$

To measure intercorrelations among two or more independent variables in all models, we conducted correlation and multicollinearity tests (results are not shown to save on space but are available upon request). The correlation matrix showed a weak relationship

between the variables (substantially lower than 0.70), and the variance inflation factor tests showed an insignificant presence of multicollinearity in all models.

3. Empirical Results

Table 5 presents the full-sample results of the probit regressions. Financial literacy was positive and strongly significant. To address the endogeneity between financial literacy and hyperbolic discounting, we present the results of the IV-probit regression in Table 6. There was a remarkable gap in the results between the probit and IV-probit regressions. The effect of financial literacy on the IV-probit regressions was negative and strongly significant. These gaps imply that there is an endogeneity bias between hyperbolic discounting and financial literacy, as expected. Moreover, we found that male gender, age, university degree, household annual income, and household financial assets are positively associated with hyperbolic discounting, while household number, unemployment, and level of risk aversion are negatively associated. We used two interaction variables such as “financial literacyXage” and “financial literacyXmale” to capture the influence of age and gender more accurately. The regression results show that both interaction variables have statistically significant positive associations with hyperbolic discounting.

Table 5. Probit regression results of hyperbolic discounting with financial literacy.

VARIABLES	PROBIT			
	Model1	Model2	Model3	Model4
financial_literacy	0.3067 *** (0.0914)	0.3080 *** (0.0915)	0.2673 *** (0.0980)	0.2619 *** (0.0982)
financial_education		−0.0277 (0.0844)	−0.0695 (0.0860)	−0.0676 (0.0860)
Male			0.0744 (0.0655)	0.0810 (0.0657)
Age			0.0024 (0.0033)	0.0018 (0.0033)
Married			0.0915 (0.1179)	0.0954 (0.1182)
Divorce			0.0837 (0.2167)	0.0946 (0.2173)
uni_degree			−0.0149 (0.0705)	−0.0192 (0.0706)
household_number			−0.0075 (0.0257)	−0.0052 (0.0258)
Children			0.3227 *** (0.1166)	0.3242 *** (0.1170)
Unemployed			−0.1033 (0.0777)	−0.1102 (0.0778)
log_of_household income			0.0500 (0.0600)	0.0442 (0.0602)
log_of_household assets			−0.0111 (0.0342)	−0.0125 (0.0343)
risk_aversion				−0.2730 *

Table 5. Cont.

VARIABLES	PROBIT			
	Model1	Model2	Model3	Model4
				(0.1532)
Constant	−0.4707 ***	−0.4675 ***	−1.482 9 *	−1.2262
	(0.0650)	(0.0657)	(0.8893)	(0.9030)
Observations	1850	1850	1850	1850
Log likelihood	−1232	−1232	−1218	−1216
Chi ² statistics	11.26	11.36	38.60	41.32
p-value	0.000794	0.00341	0.000122	8.44×10^{-5}
Robust standard errors in parentheses				

*** $p < 0.01$, * $p < 0.1$.

Table 6. Probit regression results of hyperbolic discounting using talk about finance as IV.

VARIABLES	IV-PROBIT (Talk About Finance as IV)					
	Model1	Model2	Model3	Model4	Model5	Model6
financial_literacy(IV)	−1.9154 **	−1.9549 **	−2.4947 ***	−2.4980 ***	−11.4788 ***	−2.9116 ***
	(0.8743)	(0.8830)	(0.8035)	(0.8084)	(1.7811)	(1.0794)
financial_education		0.0666	0.0129	0.0149	−0.0321	0.0416
		(0.0851)	(0.0828)	(0.0830)	(0.0627)	(0.0922)
Male			0.3164 ***	0.3237 ***	0.0952	−1.5978 ***
			(0.0739)	(0.0733)	(0.0602)	(0.5953)
Age			0.0107 ***	0.0101 ***	−0.1168 ***	0.0084 **
			(0.0035)	(0.0035)	(0.0185)	(0.0036)
Married			0.0970	0.1013	−0.0054	0.1223
			(0.1088)	(0.1087)	(0.1204)	(0.1142)
Divorce			0.0270	0.0389	0.2780 *	−0.0403
			(0.1782)	(0.1790)	(0.1593)	(0.2051)
uni_degree			0.1957 **	0.1909 **	0.1222 **	0.0577
			(0.0850)	(0.0859)	(0.0621)	(0.0656)
household_number			−0.0418 *	−0.0392	0.0066	−0.0281
			(0.0241)	(0.0243)	(0.0224)	(0.0241)
Children			0.1120	0.1141	0.0038	0.0950
			(0.1555)	(0.1562)	(0.1727)	(0.1614)
Unemployed			−0.0909	−0.0986	−0.2108 ***	−0.1121
			(0.0713)	(0.0718)	(0.0651)	(0.0740)
log_of_household income			0.2512 ***	0.2444 ***	0.1070 **	0.1827 ***
			(0.0690)	(0.0699)	(0.0481)	(0.0675)
log_of_household assets			0.0819 **	0.0802 **	−0.0113	0.0430
			(0.0397)	(0.0400)	(0.0258)	(0.0359)
risk_aversion				−0.3005 **	−0.0843	−0.3676 ***
				(0.1425)	(0.1677)	(0.1356)
Financial literacyXAge					0.2141 ***	
					(0.0314)	
Financial literacyXMale						2.9031 ***
						(0.9931)
Constant	0.9994	1.0177	−4.5551 ***	−4.2655 ***	4.6275 ***	−2.4198 ***
	(0.6175)	(0.6200)	(0.9558)	(0.9913)	(1.4477)	(0.8555)
Observations	1850	1850	1850	1850	1850	1850
Log likelihood	−1782	−1780	−1642	−1640	787.1	−1128
Chi ² statistics	4.799	4.984	111.2	123.8	412	98.67
p-value	0.0285	0.0828	0	0	0	0

Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Furthermore, Kureishi et al. [39] suggest that time discount rates decrease with age over the life cycle. Therefore, to reveal the impact of demographic category on impulsivity in more detail, we conducted a subsample analysis classified by gender and age group. Table 7 presents the results of the subsample analysis. Specifically, for the subsample classified by age group, we observed that financial literacy had a negative relationship among respondents aged between 40 and 60 and those aged over 60. Regarding the results of probit regression by gender, female samples with financial literacy were more likely to be impulsive. These results imply that the negative effect of financial literacy on impulsivity is stronger among women and increases with age.

Table 7. Probit regression results of hyperbolic discounting with financial literacy by three age groups and gender.

Variable	AGE GROUP			GENDER	
	Age < 40	40 ≤ Age < 60	Age > 60	Male	Female
financial_literacy(IV)	2.7161 (6.4591)	−2.3202 * (1.1870)	−2.3768 * (1.2955)	1.5141 (16.8705)	−2.3680 *** (0.5898)
financial_education	−0.0897 (0.3435)	−0.0592 (0.1039)	0.0766 (0.1851)	−0.0105 (0.2301)	0.0430 (0.1170)
male	−0.0834 (1.5243)	0.3845 *** (0.1118)	0.1828 (0.1494)	-	-
age	−0.0132 (0.1096)	0.0291 *** (0.0106)	−0.0264 * (0.0146)	0.0008 (0.0460)	0.0121 *** (0.0045)
married	−0.1660 (0.7151)	−0.0025 (0.1978)	0.1585 (0.1980)	−0.0233 (0.3701)	0.2556 * (0.1538)
divorce	−0.3018 (0.9759)	−0.1377 (0.2751)	−0.3372 (0.3408)	−0.2769 (1.0868)	0.0593 (0.2509)
uni_degree	−0.3849 (0.8237)	0.0486 (0.1457)	0.3716 *** (0.1328)	−0.1260 (1.4392)	0.1593 (0.1060)
household_number	0.0107 (0.1489)	−0.0077 (0.0358)	−0.1083 ** (0.0552)	0.0533 (0.2036)	−0.0525 * (0.0318)
Children	0.6310 (1.7834)	0.1256 (0.1873)	−0.1296 (0.2123)	0.3589 (1.2128)	−0.0847 (0.1697)
Unemployed	0.1036 (0.2687)	−0.0494 (0.1264)	0.0060 (0.2080)	−0.2835 (0.6502)	−0.0521 (0.0802)
log_of_household income	−0.1673 (0.5921)	0.2290 ** (0.0970)	0.2977 ** (0.1518)	−0.1297 (1.5188)	0.2290 *** (0.0773)
log_of_household assets	0.0581 (0.7555)	0.0335 (0.0610)	0.0383 (0.0933)	−0.0719 (0.6248)	0.0745 * (0.0452)
risk_aversion	−0.2599 (0.3897)	−0.4078 ** (0.2008)	−0.4143 (0.2854)	−0.5778 (0.9730)	−0.2541 (0.1993)
Constant	0.4318 (22.7973)	−4.2638 *** (1.2610)	−1.7945 (2.3626)	1.7866 (24.6023)	−4.0577 *** (1.0743)
Observations	408	944	498	877	973
Log likelihood	−346.4	−797.1	−447	−749.9	−871.4
Chi ² statistics	120.5	78.46	38.97	39.38	54.84
p-value	0	0	0.000202	9.10 × 10 ^{−5}	1.93 × 10 ^{−7}
Robust standard errors in parentheses					

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4. Discussion

This study investigated whether financial literacy affects impulsivity. We used hyperbolic discounting to measure impulsivity and employed probit regression analysis.

We find that financially literate people are less likely to be impulsive after dealing with endogeneity bias, which indicates that our hypothesis is supported. Our results are consistent with those of some studies that examine the impact of financial literacy on time preference [25,26]. However, we suggest that such effects of financial literacy can also be observed across a wide range of demographic and socioeconomic statuses. Furthermore, our subsample analyses revealed that the negative association between financial literacy and impulsivity was significant among older adults and females. To explain the results, we focus on how and why financial literacy increases over the life cycle. Some existing literature has shown that people become financially literate as they age throughout their life cycle [18]. Women are known to acquire financial knowledge, especially as they age, to prepare for their lives after their husbands' cognitive decline and death [17]. The acquisition of financial knowledge over the life cycle is motivated by future long-term maintenance of life after retirement and widowhood. Therefore, especially among females and older people, it is inferred that they will develop the ability to design their own lives in the long term from a financial perspective by gaining financial literacy, thus becoming less impulsive.

For demographic and socioeconomic variables, male respondents were more likely to be impulsive than females in the middle-aged group, which is consistent with the previous literature [40–42]. Furthermore, age is positively associated with hyperbolic discounting. This result is inconsistent with some studies that revealed that people's discounting rate is negatively correlated with age [43–45]. Given our findings that older people, particularly women, are less impulsive because they attain greater financial literacy, the positive association between age and hyperbolic discounting remains a puzzle. However, de Wit et al. [42] suggested that the negative effect of age on impulsivity varied depending on the age range of the sample. Our sample is distributed from 21 to 77 years of age and does not include child respondents. Thus, our results differ from those in the existing literature. Those who graduated from university were significantly more impulsive, which is inconsistent with the findings of Jaroni et al. [46]. This result may be due to the unique culture of university entrance examinations in Japan. Wu [47] claims that although Japan is a developed country and most families can afford higher education for their children, competition on the entrance exam is quite intense because of the limited capacity of prestigious universities. Hence, we believe that those who have taken the university entrance exam may be more competitive with others and impulsive. Alternatively, the results could be related to some issues possessed by older respondents with university education since younger respondents with a university education were not significantly more impulsive.

In addition, household financial status, such as household income and household assets, was positively significant, which is inconsistent with the findings of Green et al. [44]. One possible explanation for the results is the effects of economic conditions in the first month of 2013 when the questionnaires were distributed. On December 26, 2012, the second Abe cabinet was inaugurated, whose policy consisted of unconventional monetary policies, expansionary fiscal policies, and economic growth strategies, and they tried to stimulate the economy [48]. The stock market is strongly affected by policy [49]. We believe that this policy makes investors more impulsive.

Finally, our results revealed that risk aversion was negatively associated with impulsivity, which is consistent with the findings of van Praag and Booij [50], who found that risk aversion and the time discount rate are negatively correlated. People who tend to avoid risk may think carefully about the future.

This study had some limitations that should be considered. First, the number of observations in the subsamples was relatively small, which may have affected the results. Although we used one of the biggest behavioral surveys in Japan, the possibility of bias cannot be ruled out. Second, we excluded several observations due to missing values for

important socioeconomic variables, which forced us to compromise with the results. Third, we used data from the 2013 wave of the survey. We could not use recent data to test our hypothesis because they did not include our variables of interest. However, we believe that the results of the study would not be greatly affected because the psychological traits of people do not change much over the years. Despite these limitations, we believe that our study makes a significant contribution to the literature by providing more comprehensive evidence and a new perspective on impulsivity.

5. Conclusions

The role of financial literacy and hyperbolic discounting has been individually investigated as tools for economic and noneconomic decision making. However, it was not known how financial literacy is associated with hyperbolic discounting. This study investigated whether people's financial literacy affects their impulsivity. Our probit regression suggests that financial literacy makes people significantly less impulsive after controlling for endogeneity. In addition, our subsample analyses grouped by gender and age revealed that the effect of financial literacy on impulsivity was significant, especially among those older than 40 years and females. Our study contributes to the existing literature by providing evidence from Japan, which suggests that impulsivity is causally associated with financial literacy and provides a new perspective on impulsivity.

This study has significant implications for policymakers in Japan. As Kureishi et al. [39] claim, discount rates play a key role in economic decisions. Our study suggests that if policymakers attempt to change people's preferences in terms of impulsivity, they should consider their financial literacy. The finding that financial literacy is negatively associated with impulsive decisions supports the position of financial literacy as a rational decision-making tool. Furthermore, the findings of our study suggest that financial literacy and impulsivity should be studied together in economic and noneconomic decision making. Our study has implications at the managerial level as well. Since impulsive decisions are often observed in consumer purchasing behavior, financial literacy could be used as an instrument to reduce suboptimal economic decisions.

This study provides direction for future research on several important issues. Impulsivity should be investigated against major economic decisions such as saving, borrowing, and investing. Additionally, how impulsivity impacts health-related behavior such as exercising, drinking alcohol, maintaining health-safety measures, and gambling would be of great interest. Finally, financial literacy, age, and sex should be analyzed in depth to find a more consistent association between financial literacy and impulsivity.

Author Contributions: Conceptualization, Y.K. and M.S.R.K.; methodology, T.K., S.F., M.S.R.K. and Y.K.; software, T.K. and S.F.; validation, T.K., S.F. and Y.K.; formal analysis, T.K., S.F., M.S.R.K. and Y.K.; investigation, T.K. and S.F.; resources, Y.K.; data curation, T.K. and S.F.; writing—original draft preparation, T.K. and S.F.; writing—review and editing, M.S.R.K. and Y.K.; visualization, M.S.R.K. and Y.K.; supervision, Y.K.; project administration, Y.K.; funding acquisition, M.S.R.K. and Y.K. All authors have read and agreed to the published version of the manuscript.

Funding: This work is supported by JSPS KAKENHI, grant numbers 19K13739, 19K13684, JP23H00837, and JP23K12503. The funder had no role in the study design, data collection and analysis, preparation of the manuscript, and decision to publish.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are available upon request.

Acknowledgments: This research utilized the microdata from the Preference Parameters Study of Osaka University's 21st Century COE Program "Behavioral Macro-Dynamics Based on Surveys and Experiments," its Global COE project "Human Behavior and Socioeconomic Dynamics," and JSPS KAKENHI 15H05728 "Behavioral-Economic Analysis of Long-Run Stagnation." The authors

acknowledge these contributors to the program/projects: Yoshiro Tsutsui, Fumio Ohtake, and Shinsuke Ikeda.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Q3.

Suppose that you are to receive money from someone. You can either choose to receive the money today, or 7 days from today, but the amounts will be different. Compare the amounts and dates below in Option “A” and Option “B” and indicate which option you prefer for each of the nine choices.

Option A—Receiving Today	Option B—Receiving in 7 Days
JPY 3005	JPY 3014
JPY 3003	JPY 3297
JPY 3008	JPY 3037
JPY 3000	JPY 3000
JPY 3005	JPY 5951
JPY 3009	JPY 3068
JPY 3001	JPY 3119
JPY 3002	JPY 2996
JPY 3008	JPY 3011

Q4.

Now, suppose that you are to receive money from someone, and you can choose either to receive the money 90 days from today, or 97 days from today, but the amounts will be different. Compare the amounts and dates below in Option “A” and Option “B” and indicate which option you prefer for each of the nine choices.

Option A—Receiving in 90 Days	Option B—Receiving in 97 Days	Circle A or B	
JPY 3000	JPY 3118	A	B
JPY 3006	JPY 3000	A	B
JPY 3000	JPY 3009	A	B
JPY 3007	JPY 3301	A	B
JPY 3006	JPY 3035	A	B
JPY 3002	JPY 3005	A	B
JPY 3007	JPY 5955	A	B
JPY 3001	JPY 3001	A	B
JPY 3007	JPY 3066	A	B

Appendix B

a.

Suppose you had ¥10,000 in a savings account and the interest rate is 2% per year and you never withdraw money or receive interest payments. After 5 years, how much would you have in this account in total?

- More than ¥10,200 (correct answer)
- Exactly ¥10,200
- Less than ¥10,200
- Do not know
- Refuse to answer

b.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- More than today
- Exactly the same
- Less than today (correct answer)
- Do not know
- Refuse to answer

c.

Please indicate whether the following statement is true or false. “Buying a company stock usually provides a safer return than buying a stock mutual fund.”

- True
- False (correct answer)
- Do not know
- Refuse to answer

References

1. Khan, M.S.R.; Rabbani, N.; Kadoya, Y. Can Financial Literacy Explain Lack of Investment in Risky Assets in Japan? *Sustainability* **2021**, *13*, 12616. [\[CrossRef\]](#)
2. Khan, M.S.R.; Putthinun, P.; Watanapongvanich, S.; Yuktadatta, P.; Uddin, M.A.; Kadoya, Y. Do financial literacy and financial education influence smoking behavior in the United States? *Int. J. Environ. Res. Public Health* **2021**, *18*, 2579. [\[CrossRef\]](#) [\[PubMed\]](#)
3. Khan, M.S.R.; Rabbani, N.; Kadoya, Y. Is Financial Literacy Associated with Investment in Financial Markets in the United States? *Sustainability* **2020**, *12*, 7370. [\[CrossRef\]](#)
4. Watanapongvanich, S.; Binnagan, P.; Putthinun, P.; Khan, M.S.R.; Kadoya, Y. Financial literacy and gambling behavior: Evidence from Japan. *J. Gambl. Stud.* **2021**, *37*, 445–465. [\[CrossRef\]](#)
5. Watanapongvanich, S.; Khan, M.S.R.; Putthinun, P.; Ono, S.; Kadoya, Y. Financial literacy and gambling behavior in the United States. *J. Gambl. Stud.* **2022**, *38*, 445–463. [\[CrossRef\]](#)
6. Séguin, J.R.; Arseneault, L.; Tremblay, R.E. The contribution of “cool” and “hot” components of decision-making in adolescence: Implications for developmental psychopathology. *Cogn. Dev.* **2007**, *22*, 530–543. [\[CrossRef\]](#)
7. Kang, M.-I.; Ikeda, S. Time discounting and smoking behavior: Evidence from a panel survey*. *Health Econ.* **2013**, *23*, 1443–1464. [\[CrossRef\]](#)
8. Anokhin, A.P.; Golosheykin, S.; Grant, J.D.; Heath, A.C. Heritability of delay discounting in adolescence: A longitudinal twin study. *Behav. Genet.* **2011**, *41*, 175–183. [\[CrossRef\]](#)
9. Noguchi, R.; Shen, J. Factors affecting participation in health checkups: Evidence from Japanese survey data. *Health Policy* **2019**, *123*, 360–366. [\[CrossRef\]](#)
10. Lusardi, A.; Mitchell, O.S. The economic importance of financial literacy: Theory and evidence. *J. Econ. Lit.* **2014**, *52*, 5–44. [\[CrossRef\]](#)
11. Delavande, A.; Rohwedder, S.; Willis, R.J. *Preparation for Retirement, Financial Literacy and Cognitive Resources*; Working Paper QP 2008-190; Michigan Retirement Research Center: Ann Arbor, MI, USA, 2008. [\[CrossRef\]](#)
12. Ottaviani, C.; Vandone, D. Financial literacy, debt burden and impulsivity: A mediation analysis. *Econ. Notes* **2018**, *47*, 439–454. [\[CrossRef\]](#)
13. Hastings, J.; Mitchell, O.S. How financial literacy and impatience shape retirement wealth and investment behaviors. *J. Pension Econ. Financ.* **2020**, *19*, 1–20. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Grossman, M. On the concept of health capital and the demand for health. *J. Pol. Econ.* **1972**, *80*, 223–255. [\[CrossRef\]](#)
15. Grossman, M. The human capital model. In *Handbook of Health Economics*; Elsevier: Amsterdam, The Netherlands, 2000; Volume 1.
16. Kadoya, Y.; Khan, M.S.R. What determines financial literacy in Japan? *J. Pension Econ. Fin.* **2020**, *19*, 353–371. [\[CrossRef\]](#)
17. Hsu, J.W. Aging and strategic learning: The impact of spousal incentives on financial literacy. *J. Hum. Resour.* **2016**, *51*, 1036–1067. [\[CrossRef\]](#) [\[PubMed\]](#)
18. Jappelli, T.; Padula, M. Investment in financial literacy and saving decisions. *J. Banking Fin.* **2013**, *37*, 2779–2792. [\[CrossRef\]](#)
19. Van Rooij, M.; Lusardi, A.; Alessie, R. Financial literacy and stock market participation. *J. Financ. Econ.* **2011**, *101*, 449–472. [\[CrossRef\]](#)
20. Sekita, S. Financial literacy and retirement planning in Japan. *J. Pension Econ. Financ.* **2011**, *10*, 637–656. [\[CrossRef\]](#)
21. Lusardi, A.; Mitchell, O.S. *Financial Literacy and Retirement Planning: New Evidence from the Rand American Life Panel*; Research Paper No. WP 2007-157; Michigan Retirement Research Center: Ann Arbor, MI, USA, 2007. [\[CrossRef\]](#)
22. Bernheim, B.D. Do households appreciate their financial vulnerabilities? An analysis of actions, perceptions, and public policy. In *Tax Policy and Economic Growth*; American Council for Capital Formation: Washington, DC, USA, 1995; pp. 1–30.
23. Zhang, L. Saving and retirement behavior under quasi-hyperbolic discounting. *J. Econ.* **2013**, *109*, 57–71. [\[CrossRef\]](#)
24. Ikeda, S.; Kang, M.; Ohtake, F. Hyperbolic discounting, the sign effect, and the body mass index. *J. Health Econ.* **2010**, *29*, 268–284. [\[CrossRef\]](#)

25. Mudzingiri, C.; Mwamba, J.W.M.; Keyser, J.N.; Bara, A. Indecisiveness on risk preference and time preference choices. Does financial literacy Matter? *Cogent Psychol.* **2019**, *6*, 1647817. [\[CrossRef\]](#)
26. Mudzingiri, C. The impact of financial literacy on risk seeking and patient attitudes of university students. *Dev. S. Afr.* **2021**, *38*, 845–861. [\[CrossRef\]](#)
27. Lusardi, A.; Mitchell, O.S. Planning and financial literacy: How do women fare? *Am. Econ. Rev.* **2008**, *98*, 413–417. [\[CrossRef\]](#)
28. Lusardi, A.; Mitchell, O.S. Financial literacy around the world: An overview. *J. Pension Econ. Financ.* **2011**, *10*, 497–508. [\[CrossRef\]](#)
29. Lusardi, A.; Mitchell, O.S.; Curto, V. Financial literacy among the young. *J. Con. Aff.* **2010**, *44*, 358–380. [\[CrossRef\]](#)
30. Lal, S.; Nguyen, T.X.T.; Sulemana, A.-S.; Khan, M.S.R.; Kadoya, Y. Does financial literacy influence preventive health check-up behavior in Japan? A cross-sectional study. *BMC Public Health* **2022**, *22*, 1704. [\[CrossRef\]](#) [\[PubMed\]](#)
31. Nguyen, T.X.T.; Lal, S.; Abdul-Salam, S.; Khan, M.S.R.; Kadoya, Y. Financial literacy, financial education, and cancer screening behavior: Evidence from Japan. *Int. J. Environ. Res. Public Health* **2022**, *19*, 4457. [\[CrossRef\]](#)
32. Klapper, L.; Panos, G.A. Financial literacy and retirement planning: The Russian case. *J. Pension Econ. Fin.* **2011**, *10*, 599–618. [\[CrossRef\]](#)
33. Samuelson, P.A. A note on measurement of utility. *Rev. Econ. Stud.* **1937**, *4*, 155. [\[CrossRef\]](#)
34. Story, G.W.; Vlaev, I.; Seymour, B.; Darzi, A.; Dolan, R.J. Does temporal discounting explain unhealthy behavior? A systematic review and reinforcement learning perspective. *Front. Behav. Neurosci.* **2014**, *8*, 76. [\[CrossRef\]](#)
35. Green, L.; Myerson, J. Exponential versus Hyperbolic Discounting of Delayed Outcomes: Risk and Waiting Time. *Am. Zool.* **1996**, *36*, 496–505. [\[CrossRef\]](#)
36. Jorgensen, B.L.; Savla, J. Financial literacy of young adults: The importance of parental socialization. *Fam. Relat.* **2010**, *59*, 465–478. [\[CrossRef\]](#)
37. Van Campenhout, G. Revaluing the role of parents as financial socialization agents in youth financial literacy programs. *J. Con. Aff.* **2015**, *49*, 186–222. [\[CrossRef\]](#)
38. Norvilitis, J.M.; MacLean, M.G. The role of parents in college students' financial behaviors and attitudes. *J. Econ. Psychol.* **2010**, *31*, 55–63. [\[CrossRef\]](#)
39. Kureishi, W.; Paule-Paludkiewicz, H.; Tsujiyama, H.; Wakabayashi, M. Time preferences over the life cycle and household saving puzzles. *J. Monet. Econ.* **2021**, *124*, 123–139. [\[CrossRef\]](#)
40. Stoltenberg, S.F.; Batien, B.D.; Birgenheir, D.G. Does gender moderate associations among impulsivity and health-risk behaviors? *Addict. Behav.* **2008**, *33*, 252–265. [\[CrossRef\]](#)
41. Kirby, K.N.; Maraković, N.N. Modeling myopic decisions: Evidence for hyperbolic delay discounting within subjects and amounts. *Organ. Behav. Hum. Decis. Processes* **1995**, *64*, 22–30. [\[CrossRef\]](#)
42. de Wit, H.; Flory, J.D.; Acheson, A.; McCloskey, M.; Manuck, S.B. IQ and nonplanning impulsivity are independently associated with delay discounting in middle-aged adults. *Pers. Individ. Dif.* **2007**, *42*, 111–121. [\[CrossRef\]](#)
43. Green, L.; Fry, A.F.; Myerson, J. Discounting of delayed rewards: A life-span comparison. *Psychol. Sci.* **1994**, *5*, 33–36. [\[CrossRef\]](#)
44. Green, L.; Myerson, J.; Lichtman, D.; Rosen, S.; Fry, A. Temporal discounting in choice between delayed rewards: The role of age and income. *Psychol. Aging* **1996**, *11*, 79–84. [\[CrossRef\]](#) [\[PubMed\]](#)
45. Petry, N.M. Delay discounting of money and alcohol in actively using alcoholics, currently abstinent alcoholics, and controls. *Psychopharmacology* **2001**, *154*, 243–250. [\[CrossRef\]](#)
46. Jaroni, J.L.; Wright, S.M.; Lerman, C.; Epstein, L.H. Relationship between education and delay discounting in smokers. *Addict. Behav.* **2004**, *29*, 1171–1175. [\[CrossRef\]](#) [\[PubMed\]](#)
47. Wu, L.-e.E.T. Japanese university entrance examination problems in mathematics. *Am. Educ. Prof. J. Am. Fed. Teach.* **1993**, *17*, 25–35.
48. Fukuda, S.-I. Abenomics: Why was it so successful in changing market expectations? *J. Jpn. Int. Econ.* **2015**, *37*, 1–20. [\[CrossRef\]](#)
49. Ito, T. An assessment of Abenomics: Evolution and achievements. *Asian Econ. Policy Rev.* **2021**, *16*, 190–219. [\[CrossRef\]](#)
50. van Praag, B.; Booij, A.S. Risk aversion and the subjective time discount rate: A joint approach. *SSRN Electron. J.* **2003**, 396563. [\[CrossRef\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.